Jules De Meo

C and C++ developer, building heavy machines in code

Perpignan, France · Relocating to Czechia, November 2026 · julesdemeo05@gmail.com · +33 7 84 77 02 11 · shrrimp.github.io · coachforce.fr

PROFILE

I like understanding how things work all the way down, whether it is a hydraulic arm, an overdrive circuit or a heavy squat. That curiosity is behind most of what I build: a physics library for heavy machines, a game where people design their own, guitar pedals, and *CoachForce*, the coaching platform I wrote for my own sport and still run. I learned to program at 42, where every project is reviewed by peers, and everything since then I have built alone. What I want next is a team to build with.

SELECTED PROJECTS

Heavy Feather, open-source physics for heavy machinery

2026 to present

C++20, CMake, no dependencies, MIT licence · github.com/shrrimp/heavy-feather

A C++ library for machines whose parts push hard on each other: linkages, hydraulic arms, cranes, presses, drivetrains.

Each machine is one Featherstone articulation, so joints stay exact under load and lopsided masses stay true. Getting there took a lot of reading and a lot of mechanisms exploding on screen.

Released as v0.1 with motors, pistons and friction. Before I trust a change I run four solver implementations on the same scenes, and I check that every test can actually fail.

Heavy Metal, a heavy-machinery engineering sandbox

2026 to present, in development

C++20, Vulkan ray tracing, GLSL, Linux · preview videos at shrrimp.github.io

Players design any heavy machine they like out of voxels, and the physics decides whether it works. Heavy Feather began as its physics prototype.

I wrote the renderer by hand too: fully ray-traced with global illumination, around 3 ms for primary visibility and 15 ms for a full 1080p frame on a mid-range GPU.

Guitar pedals, from simulation to bench

2026 to present

ngspice, CircuiTikz, LaTeX, Git

I play guitar, so I started designing my own effects: a tunable overdrive and distortion, simulated in ngspice, then soldered and debugged with an oscilloscope until it sounded right, with every schematic change tracked in Git.

EXPERIENCE

Founder and developer, *CoachForce*

Sept. 2025 to present

coachforce.fr · 42 professional experience, Apr. to Oct. 2026 · TypeScript, Next.js, Prisma, PostgreSQL, Stripe Connect

Born from my own sport: a multi-tenant platform where strength coaches write programs, follow their athletes and get paid, designed, built and run by me alone.

The analytics engine turns raw training logs into the numbers a coach actually looks at.

Billing on Stripe Connect with connected-account onboarding and KYC, and file storage split between AWS S3 and Cloudflare R2 to keep costs down.

I spent as much time talking with around thirty coaches as writing code, and the roadmap changed because of it.

EDUCATION

Software engineering, *42 Perpignan*

Oct. 2023 to Oct. 2026

Curriculum completed · around 2,000 hours of project work, no lectures, no teachers

Systems programming in C and C++: memory management, processes, threads, sockets, parsing, and parts of the C standard library rewritten from scratch.

Every project defended in front of other students, under strict rules on style, leaks and crashes. That is where I learned to explain my code, and to hear what was wrong with it.

Baccalauréat général, *French national diploma*

2022

Majors in mathematics, computer science, and physics and chemistry

TECHNICAL

Languages	C, C++ (98 to 20), Python, Bash, TypeScript, SQL
Physics	Articulated-body dynamics (Featherstone), contacts with friction, closed-loop mechanisms, differential testing of solvers
Systems	Multi-threading, memory management, processes and pipes, sockets, CPU and GPU profiling
Graphics	Vulkan ray tracing, GLSL, acceleration structures, GPU acceleration
Web and data	React, Next.js, Node, Prisma, PostgreSQL, REST APIs, Stripe, Vercel
Tooling	Linux only, Arch as daily driver, Git, Docker, CMake, ngspice

LANGUAGES

English every day, as comfortable as my native French. Czech: beginner, and learning.

BEYOND WORK

National-level powerlifter, 74 kg class, French federation, and the reason CoachForce exists. I am also building my own electric guitar from scratch.